

Nifty 100 Financial Intelligence Platform

Analyst Guide

Project Overview

This platform provides comprehensive financial analysis of the Nifty 100 index. It includes ETL pipelines, financial ratio computations, screening, peer comparison, sector analysis, and a Streamlit dashboard.

Deliverables Summary

- D-01: SQLite database (nifty100.db)
- D-02: Load audit CSV
- D-03: Validation failures CSV
- D-04: SQL notebooks
- D-05: Financial ratios table
- D-06: Capital allocation CSV
- D-07: Screener output Excel
- D-08: Screener config YAML
- D-09: Peer comparison Excel
- D-10: Radar charts (PNG)
- D-11: Streamlit Dashboard (8 screens)
- D-12: Valuation summary Excel
- D-13: Cash flow intelligence Excel
- D-14: Pros/cons generated CSV
- D-15: Analysis parsed CSV
- D-16: Company tearsheets (PDF)
- D-17: Sector reports (PDF)
- D-18: Portfolio summary PDF
- D-19: Cluster labels CSV
- D-20: FastAPI server (16 endpoints)
- D-21: Pytest report HTML
- D-22: Analyst guide (this document)
- D-23: Acceptance checklist PDF

Key Commands

make load - Load all Excel files into nifty100.db
make ratios - Generate financial_ratios table
make test - Run all pytest tests
make report - Generate tearsheets, sector reports, portfolio summary
make dashboard - Launch Streamlit Dashboard on localhost:8501
make api - Launch FastAPI server on localhost:8000
make clean - Remove cache and test artifacts

FastAPI Endpoints (D-20)

GET /api/v1/health - Health check
GET /api/v1/companies - List all companies (filterable)
GET /api/v1/companies/{ticker} - Company profile
GET /api/v1/companies/{ticker}/pl - P&L; history
GET /api/v1/companies/{ticker}/bs - Balance sheet history
GET /api/v1/companies/{ticker}/cashflow - Cash flow history
GET /api/v1/companies/{ticker}/ratios - Financial ratios
GET /api/v1/companies/{ticker}/tearsheet - Download PDF tearsheet
GET /api/v1/screener - Run screener with filters
GET /api/v1/portfolio/stats - Portfolio percentile stats
GET /api/v1/sectors - All sectors with stats
GET /api/v1/sectors/{sector}/companies - Companies in sector
GET /api/v1/peers/{group_name} - Peer group members + benchmark
GET /api/v1/companies/{ticker}/peers/compare - Peer radar data
GET /api/v1/market-cap/{ticker} - Historical PE, PB, EV/EBITDA
GET /api/v1/companies/{ticker}/documents - Annual report links

Dashboard Screens (D-11)

1. Home - Overview KPIs, sector breakdown, quality score leaders
2. Company Profile - Deep-dive into individual companies
3. Screener - 6 preset screeners + 10 custom metric sliders
4. Peer Comparison - Radar chart vs peer group averages
5. Trend Analysis - Multi-metric 10-year trend charts
6. Sector Analysis - Bubble chart and sector median KPIs
7. Capital Allocation - Treemap of 9 allocation patterns
8. Annual Reports - BSE annual report links by year

Important Coding Rules

1. All Excel files use `pd.read_excel(path, header=1)`
2. `company_id` normalized (trim + uppercase) before joins
3. All monetary values stored in INR Crore
4. D/E screener skips Financials sector
5. Negative base year CAGR returns TURNAROUND
6. Interest Expense = 0 shows Debt Free
7. Simulated data labeled in dashboards/reports
8. Run make test before every git commit

Database Schema

companies, profitandloss, balancesheet, cashflow, analysis, documents, prosandcons, sectors, stock_prices, market_cap, financial_ratios, peer_groups, peer_percentiles

Technical Stack

Python 3.11+, FastAPI, Streamlit, SQLite, pandas, plotly, matplotlib, scikit-learn, pytest